

The empirical recurrence for OEIS A299461, recovered from its tail

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Abstract

OEIS A299461 records “Empirical recurrence of order 69” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 72$. Its last coefficient is $c_{69} = -3584 \neq 0$, so no further tail satisfies anything shorter; any order-69 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299461 is “Number of $n \times 4$ 0..1 arrays with every element equal to 1, 2, 3, 5, 6 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

2, 52, 226, 1013, 5294, 27639, 139226, 713421, ...

The entry states:

Empirical recurrence of order 69 (see link above)

As of the “Last modified” line on the live entry (Feb 10 2018 10:44:15, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 69: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 69.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 3$ on, it returns order 69 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{69} c_i a(n-i),$$

c_1	10	c_{19}	10011468	c_{37}	−1354869141	c_{55}	26144203
c_2	−36	c_{20}	−2639465	c_{38}	1098076287	c_{56}	−116470479
c_3	124	c_{21}	−614673	c_{39}	756444016	c_{57}	111959512
c_4	−574	c_{22}	17719570	c_{40}	−485894968	c_{58}	20112822
c_5	1602	c_{23}	−54989464	c_{41}	−214426742	c_{59}	−17481720
c_6	−3870	c_{24}	80758957	c_{42}	−1472523191	c_{60}	−7759762
c_7	12788	c_{25}	−173045122	c_{43}	429846770	c_{61}	3758142
c_8	−29283	c_{26}	263863511	c_{44}	−82838488	c_{62}	−3667730
c_9	58522	c_{27}	−337625440	c_{45}	1973411831	c_{63}	−380700
c_{10}	−155131	c_{28}	553297104	c_{46}	−959453298	c_{64}	1279976
c_{11}	297804	c_{29}	−699251480	c_{47}	−14132449	c_{65}	193032
c_{12}	−528406	c_{30}	958349976	c_{48}	−378801408	c_{66}	−192948
c_{13}	1157035	c_{31}	−1273104543	c_{49}	−327888362	c_{67}	10856
c_{14}	−1820532	c_{32}	1438049832	c_{50}	244810875	c_{68}	9088
c_{15}	2916692	c_{33}	−1852860246	c_{51}	386965023	c_{69}	−3584
c_{16}	−5151702	c_{34}	2120286945	c_{52}	248138194		
c_{17}	5908187	c_{35}	−2369125330	c_{53}	−475323885		
c_{18}	−8078569	c_{36}	1908602571	c_{54}	54187717		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 72$, and it is the recurrence OEIS A299461 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{69} - c_1 x^{68} - \dots - c_{69}$. Its constant term is $-c_{69} = 3584$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 69 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 69, so $\deg q = 69$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 69, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 69, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -3584 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-69 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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